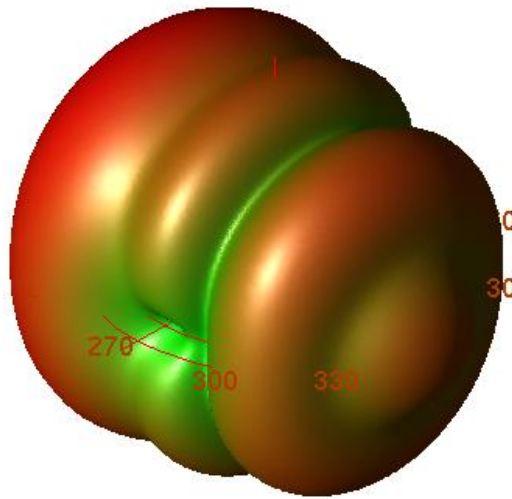


EE351 Project: Study of Antennas via Keysight's Advanced Design System Software



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EE351

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Abstract

In this report, different types of antenna models are built and simulated using Keysight's Advanced Design System, or ADS. The hertzian Dipole, long filamentary antenna, antenna array, and path antenna are simulated using ADS software, and they will be compared with the equations that we used for EE 351 course using MATLAB.

Introduction

Hertzian Dipole is a microcurrent element which entire length is significantly shorter than its operating wavelength. Since its length is short, it can be assumed that the current distribution across the antenna is uniform. Hertzian dipole is used as the most fundamental building block when calculating the electromagnetic fields of complex antennas.

Arbitrary length filamentary antenna varies its current dependence on the position of the antenna unlike a hertz dipole. The current at both ends should be zero, so it is estimated that it has a standing wave pattern like a transmission line with an open circuit. Half-wave dipoles with half-wavelength are commonly used because their radiation resistance reaches about seventy-three ohms. It makes it easy to match the commercial coaxial cables.

A single antenna element alone has limitation in increasing the directionality of concentrating radio waves in a specific direction. Thus, multiple radiation elements are configured, and it is called antenna array. Same amplitude of current is supplied to each element, and they are placed by a constant gap and gradual phase shift. Depending on the combination of distance between each element and phase difference, broadside array or end-fire array can be induced.

Patch antenna is easily found in the mobile phone market. It is easy to printed onto a circuit board, and it can be fabricated into the circuit board with the low cost. Unlike other

antenna models, it has to add the ground plane. Ground plane is usually a perfect conductor, or PEC, and dielectric substrate is above the ground plane. And the patch is located on top of the substrate.

Results

Hertzian Dipole

To draw the layout for hertzian dipole, Appendix F in the [1] was used. Each antenna trace has length of one hundred mil and width of forty mil. The thickness of the antenna is forty mil. Antenna itself is perfect conductor, and it is surrounded by air. Figure 1 is the layout of the hertzian dipole model.

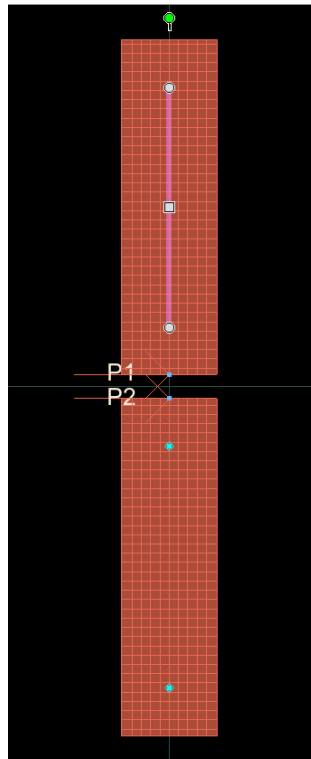


Figure 1. Hertzian Dipole Model Layout.

After drawing the layout, the simulation was run. The result of the simulation is in figure 2 and 3. Figure 2 is the antenna pattern for the hertzian dipole, and figure 3 is the current

distribution for the hertzian dipole. A uniform current distribution can be checked with this simulation results.

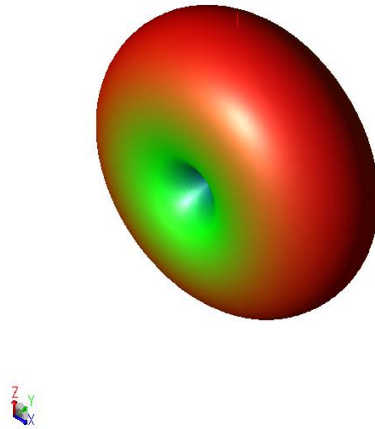


Figure 2. Hertzian Dipole Antenna Pattern.

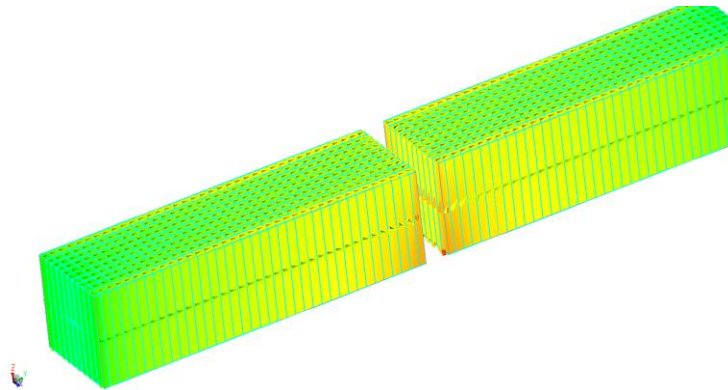


Figure 3. Hertzian Dipole Current Distribution.

Figure 4 shows the G_d of the hertzian dipole, and it is identical to the directivity gain that is shown in the lecture notes. To check its similarity or difference between ADS simulation and the theory that is taught in the EE 351 class, the data from simulation is extracted and compared with the theory equation that is used in the EE 351 courses.

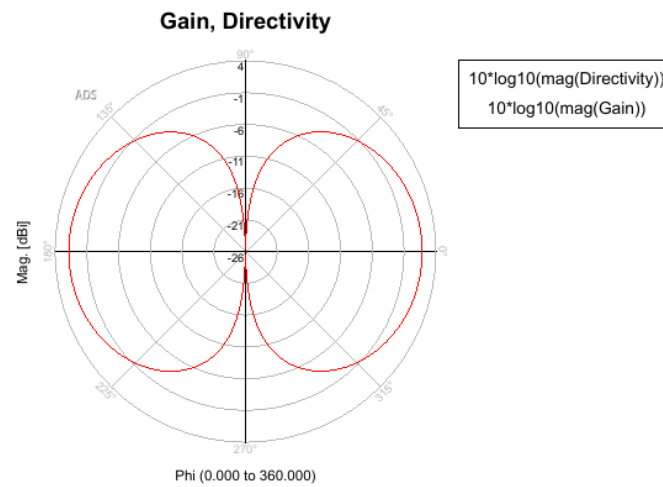


Figure 4. Hertzian Dipole Directivity.

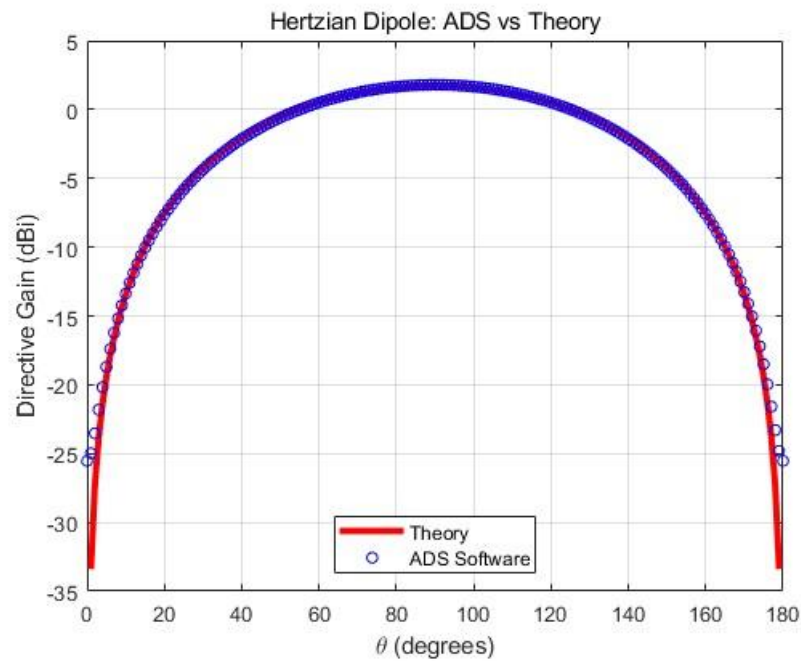


Figure 5. ADS simulation Vs Theory.

As can be found in figure 5, the simulation results matched the theory that is taught in the EE 351 course. Figure 6 is the MATLAB code that is used for generating figure 5.

```

1  clear; clf;
2
3  ADS_data = load('01_Hertzian_Qaudrant1Export2Matlab_ASCII.txt');
4
5  % Convert Phi to Theta
6  phi_ads_deg = ADS_data(:,1);
7  theta_EE351 = phi_ads_deg - 90;
8
9  Gd_dB_ADS = ADS_data(:,2);
10
11 % filter the data that is less then 0 or more than 180
12 for counter = 1:length(theta_EE351)
13     if theta_EE351(counter) < 0
14         theta_EE351(counter) = nan;
15         Gd_dB_ADS(counter) = nan;
16     elseif theta_EE351(counter) > 180
17         theta_EE351(counter) = nan;
18         Gd_dB_ADS(counter) = nan;
19     end
20 end
21
22 % Hertzian Dipole's Directive Gain)
23 % Gd = 1.5 * sin^2(theta)
24 Gd_theory_HD = 1.5 * (sind(theta_EE351)).^2;
25 Gd_theory_HD_dB = 10 * log10(Gd_theory_HD);
26
27 % plot
28 plot(theta_EE351, Gd_theory_HD_dB, 'r-', 'Linewidth', 3); hold on;
29 plot(theta_EE351, Gd_dB_ADS, 'bo', 'MarkerSize', 5);
30
31 axis();
32 xlabel('\theta (degrees)');
33 ylabel('Directive Gain (dBi)');
34 title('Hertzian Dipole: ADS vs Theory');
35 grid on;
36 legend('Theory', 'ADS Software', 'Location', 'south');
37

```

Figure 6. MATLAB code for generating figure 5.

Long Filamentary Antenna

To generate the long filamentary antenna, the length of the hertzian dipole should be increased up to personalized scaling factor, k_p . k_p can be calculated with this equation:

$$k_p = 13 * \xi$$

$$k_p = 13 * 370 = 4810$$

My personal k_p is 4810, so the full length of the antenna is 4810 mil. Then, each length of antenna is 2405 mil.

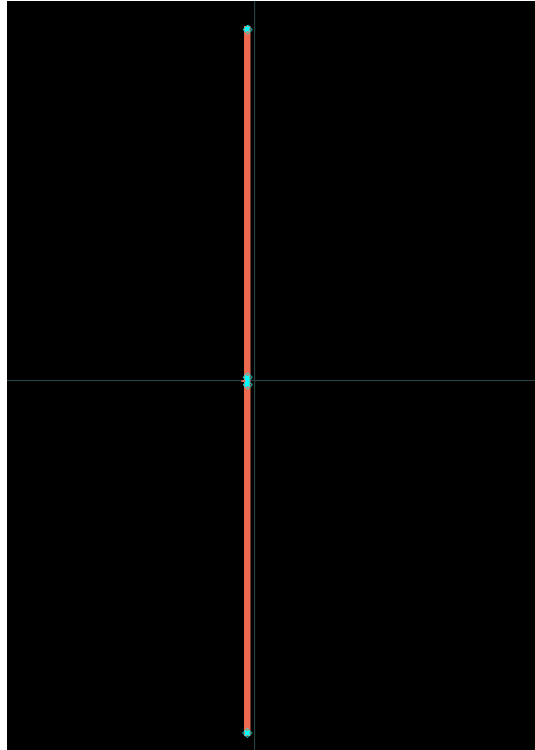


Figure 7. Long Filamentary Antenna Layout

Long antenna produced a tubed apple shape of antenna pattern. Figure 8 shows the antenna pattern of the long filamentary antenna. Compared to the hertzian dipole, it has more lips between the end of the “apple.”

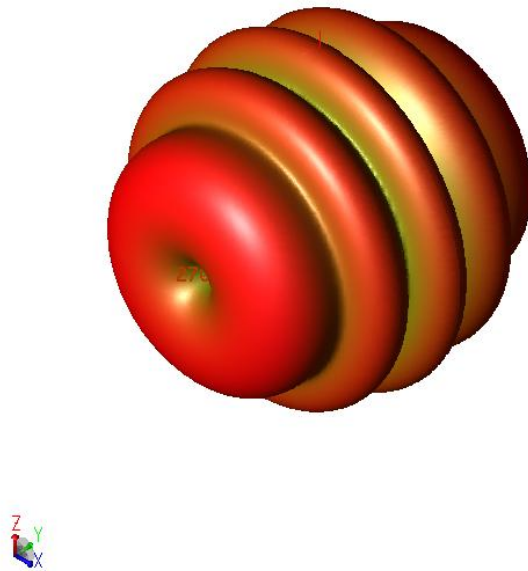


Figure 8 Long Filamentary Antenna Pattern.

The current distribution of the long filamentary antenna is not uniform. Figure 9 is the full view of the current distribution, and figure 10 is the close-up shot of the current distribution. Unlike the hertzian dipole's current distribution, it is not uniform.

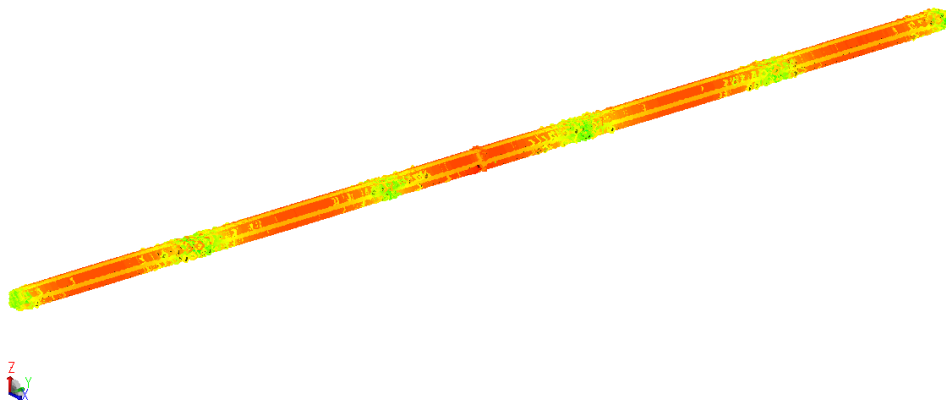


Figure 9. Long Filamentary Antenna Current Distribution

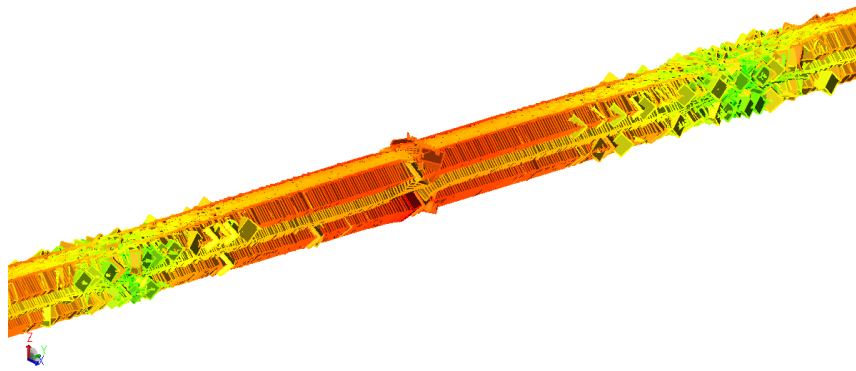


Figure 10. Close-up Long Filamentary Antenna Current Distribution

Figure 11 is the directivity gain of the ADS simulation. As we learned in the EE 351 course, the shape of directivity is not just donut shape, but it has many lobes. Figure 12 is the comparison between the simulation results and the theory from EE 351 course, and they are almost identical. Figure 13 is the MATLAB code to generate figure 12.

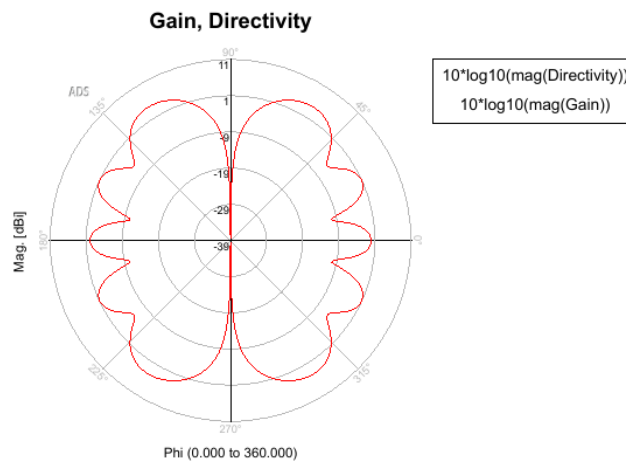


Figure 11. Long Filamentary Antenna Directivity

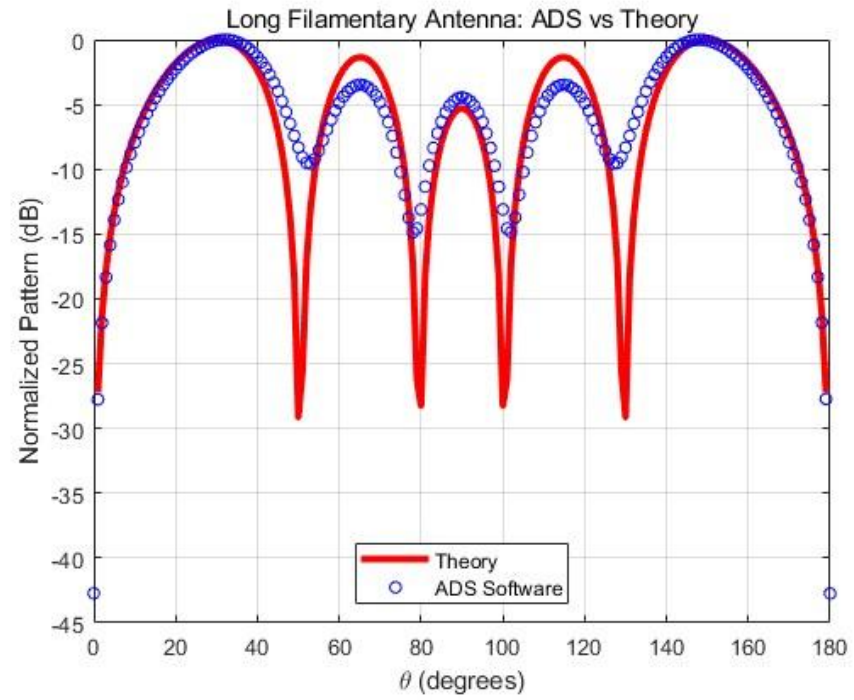


Figure 12. ADS simulation Vs Theory.

```

1  clear; clf;
2  xi = 370;
3  % calculate the Kp
4  L_mil = 13 * xi;
5  lambda_mil = 1969;
6  beta = 2 * pi / lambda_mil;
7
8
9  ADS_data = load('02_Long_Quadrant1Export2Matlab_ASCII.txt');
10 % convert the coordinate
11 phi_ads_deg = ADS_data(:,1);
12 theta_EE351 = phi_ads_deg - 90;
13
14 % ADS Directive Gain data
15 Gd_dB_ADS = ADS_data(:,2);
16
17 % DATA filtering
18 for counter = 1:length(theta_EE351)
19     if theta_EE351(counter) < 0
20         theta_EE351(counter) = nan;
21         Gd_dB_ADS(counter) = nan;
22     elseif theta_EE351(counter) > 180
23         theta_EE351(counter) = nan;
24         Gd_dB_ADS(counter) = nan;
25     end
26 end
27
28 % Long Filamentary Antenna's Normalized E-field
29 theta_rad = theta_EE351 * pi / 180;
30 numerator = cos((beta * L_mil / 2) .* cos(theta_rad)) - cos(beta * L_mil / 2);
31 denominator = sin(theta_rad);
32 denominator(denominator == 0) = eps;
33
34 F_theta = numerator ./ denominator; % Far zone
35 Power_pattern_theory = F_theta.^2;
36
37 % dB covert and normalization.
38 Power_theory_dB = 10 * log10(Power_pattern_theory);
39 Power_theory_dB_norm = Power_theory_dB - max(Power_theory_dB);
40
41 Gd_dB_ADS_norm = Gd_dB_ADS - max(Gd_dB_ADS);
42
43 % Plot
44 plot(theta_EE351, Power_theory_dB_norm, 'r-', 'Linewidth', 3); hold on;
45 plot(theta_EE351, Gd_dB_ADS_norm, 'bo', 'MarkerSize', 5);
46
47 % axis
48 axis();
49 xlabel('\theta (degrees)');
50 ylabel('Normalized Pattern (dB)');
51 title('Long Filamentary Antenna: ADS vs Theory');
52 grid on;
53 legend('Theory', 'ADS Software', 'Location', 'south');

```

Figure 13. MATLAB code for generating figure 23.

Antenna Array

Hertzian Dipole is used to generate the antenna array. N is set to four, hence four hertzian dipole is placed with the d distance. d is set to $\frac{\lambda}{2}$. Since the operating frequency is six giga hertz,

λ is 1969 mil. Thus, half λ is equal to 985 mil. Each hertzian dipole is separated by 985 mil.

Figure 14 is the layout of the antenna array.

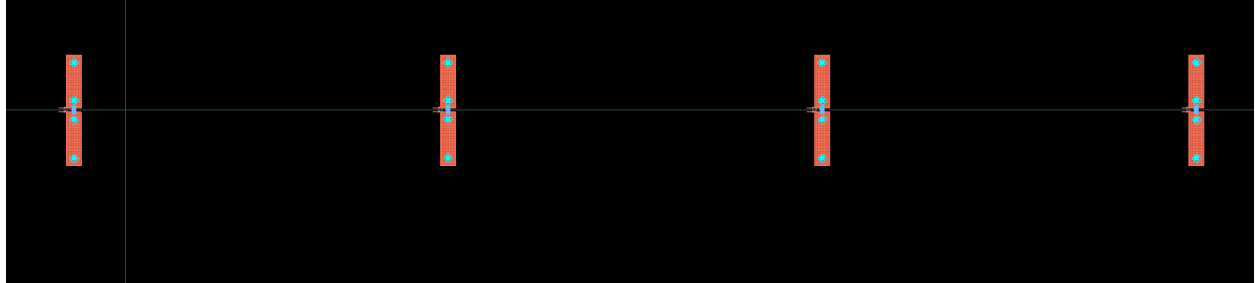


Figure 14. Antenna Array Layout.

Gradual phase shift, α , is equal to βd , and α can be calculated.

$$\alpha = \beta d$$

$$\alpha = \frac{2\pi}{\lambda} * \frac{\lambda}{2}$$

$$\alpha = \pi = 180 \text{ degree}$$

The first element of the array's phase shift is set to zero. The second element is set to 180 degrees. The third element is set to 360 degrees, or zero. Lastly, the fourth element is set to 180 degrees. All the elements' amplitude is set to 1. Since it has a gradual phase shift, it is end-fire array. The antenna pattern results of antenna array is in figure 15. It has two main lobes and four minor lobes.

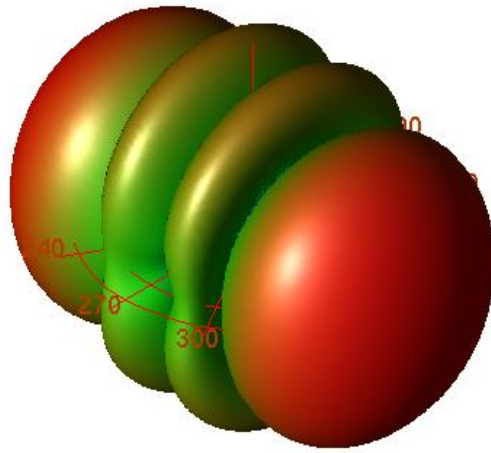


Figure 15. Antenna Array Pattern.

Figure 16 is the directivity of end-fire antenna array. Ten step graphical technique is used to verify the ADS simulation if the simulation fits with the theory from EE 351 course. Since α and the radius is π , the end point of the circle should be meet the origin of array factor graph. The other end point is met 2π , which is approximately 6.28. Figure 17 is the hand drew antenna array directivity from the array factor. Both simulated directivity and hand drew directivity is almost identical to each other.

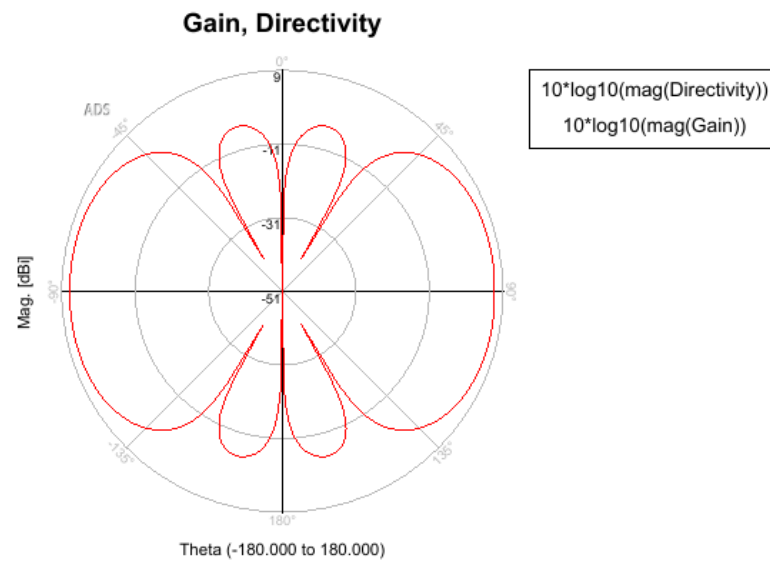


Figure 16. Antenna Array Directivity.

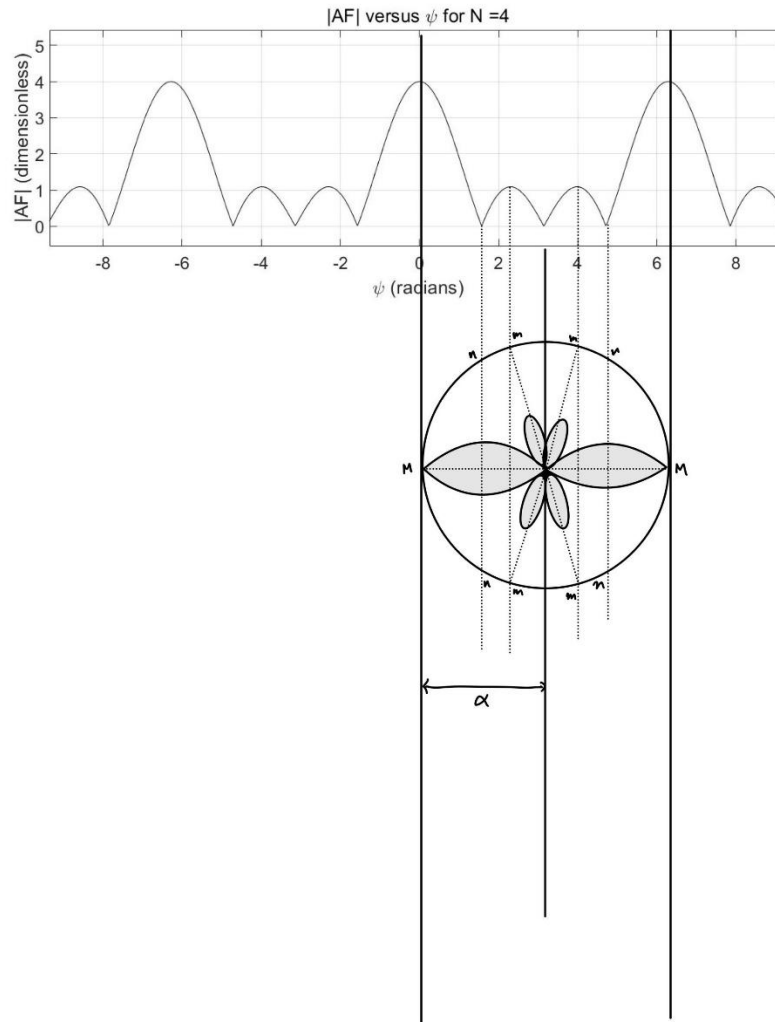


Figure 17. 10 Step Graphical Technique

Patch Antenna

Eight steps of modeling patch antenna in [1] is used to draw the layout of patch antenna. The width of the trace is 500 mil, and the length of the trace is also 500 mil. It is connected to pin one, which negative port is connected to ground. Ground plane is set to PEC, and the length of dielectric between substrate to ground plane is set to 10 mil. Patch antenna is taking low cost

to integrate into circuit boards, so it is commonly use in the mobile phone markets. The layout of patch antenna is in Figure 18.

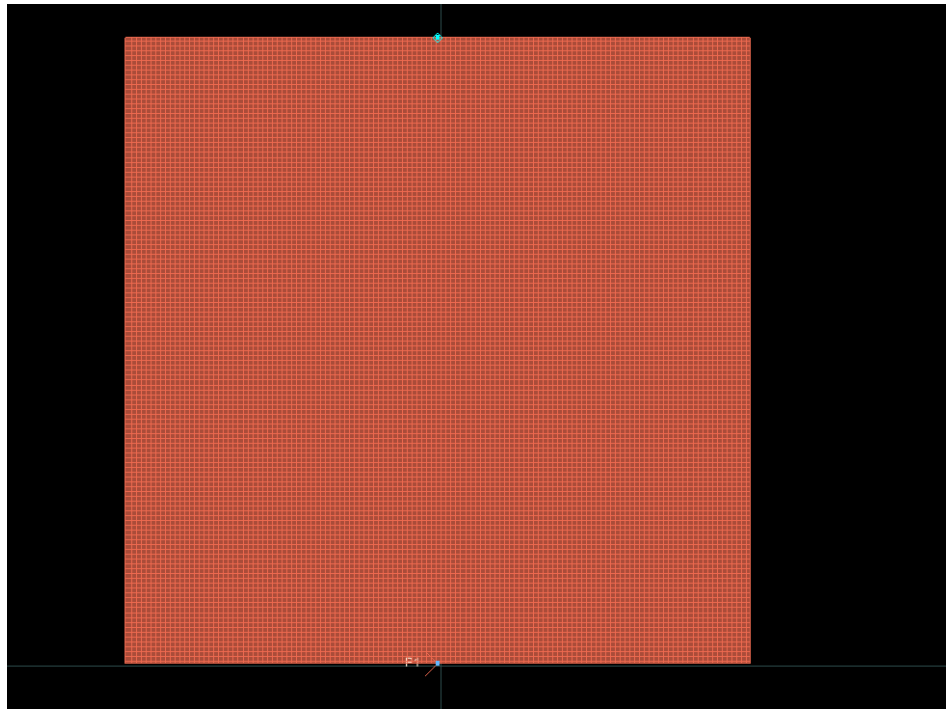


Figure 18. Patch Antenna Layout.

Since it has the ground plane at the Bottom, the antenna pattern did not show like the donut or apple shape. Antenna pattern of the patch antenna looks like lollipop hat has flat bottom and spherical top. Patch antenna's antenna pattern is in figure 19.

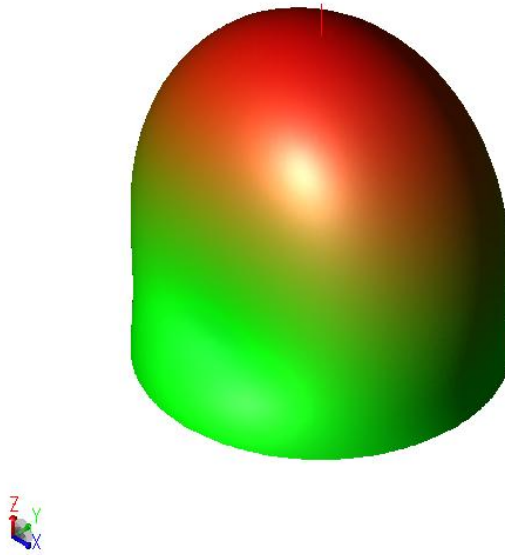


Figure 19. Patch Antenna Pattern

The current distribution of the patch antenna is stable throughout the surface. Unlike the arbitrary long filamentary antenna, the result of simulation seems uniform. The current distribution simulation is in Figure 20, and its close-up snapshot is in Figure 21.

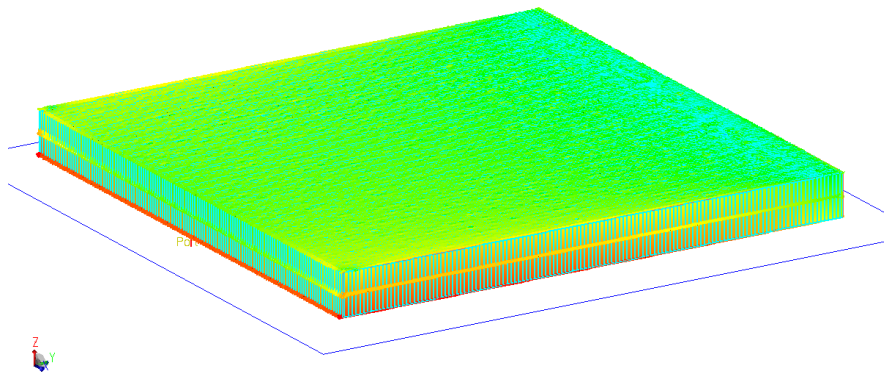


Figure 20. Patch Antenna Current Distribution

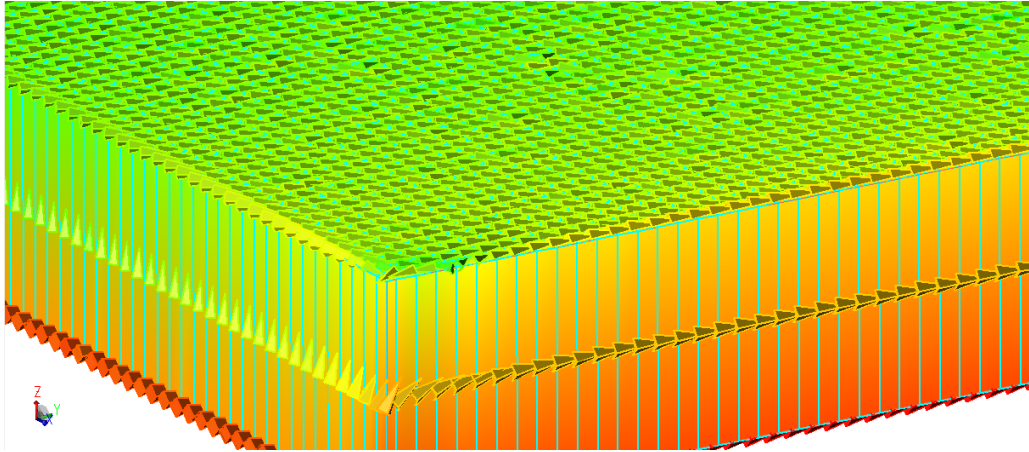


Figure 21. Close-up Patch Antenna Current Distribution

Contemporary Issue

The well-known issue for faster communication is short coverage. We are living with 5G technology, and soon, we will experience 6G. For 6G, broader bandwidth is essential for faster data transportation. It uses not only mmWave such as 28 GHz or 38 GHz, it uses terabits-per-second communication for 6G, which might be faster than the optical networks. To solve this problem, companies are researching to make a device that can enable more antenna and higher transmit capabilities. [8] talks about the FWA and sustainability of FR2.

Discussion

The result of hertzian dipole simulations meets the theory that we learned in the EE 351 course. The simulation curve and the theory curve are identical to each other in the MATLAB result, figure 5. The result of long filamentary antenna is almost identical to the theory. One reason it is not perfectly matched should be the gap between the two antenna traces. Even though this error, the MATLAB result, figure 12, can be said to be close enough. The directivity of antenna array and the ten step graphical technique is similar to each other.

Conclusions

The simulation results of hertzian dipole, long filamentary antenna, and antenna array are similar to the theory that we learned through the EE 351 course. Thus, the simulation results and theory have met the agreements.

References

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- [2] P. Pedrow, “EE351_pages_142_147.pdf”, EE351, Washington State University, Pullman, WA, 2026
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- [8] BLOG | Samsung Research, “BLOG | Samsung Research,” *BLOG | Samsung Research*, 2026. <https://research.samsung.com/blog/6G-FWA-Broadband-Beyond-Smartphones> (accessed May 02, 2026).